

teacher reference

This is your **concept + vocabulary anchor** for the electronics programme. The children's book is self-serving – they explore, experiment, complete circuits, build, solve and present on their own. This guide is for **you**: the **concept** behind each machine, the exact **words** to teach, the **experiment** and the **answers**, so every teacher covers the same syllabus.

Each machine gives you: 💡 **the idea** · 🔑 **the words to teach** · 🖋️ **the experiment + answers** (5-8 and 8-12) · ⚡ **the circuit exercise + answer** · 🌐 **the real-life link**.

Depth to aim for: words like *current*, *charge*, *voltage* are for **familiarity, not mastery** – they appear as "did you know?" facts and come back on every machine. Keep the simple words consistent everywhere.

The bands: 5-8 completes circuits and answers by circling. 8-12 **designs** its own circuits and draws them as schematics. There is no multimeter in the kit – 8-12 "measuring" means structured comparison, not voltage numbers.

before you start

safety – what to teach

The child's book has its own safety page. These are the points to say aloud and enforce.



The battery blocks in the kit are **safe** to handle.



Mains power is **not**. Children must never put anything into a wall socket – say this out loud on day one.



No wet hands near switches, plugs or wires.



Wires wear a plastic coat – an **insulator**. Retire any wire with the coat torn.



Anything hot or burnt-smelling: stop, disconnect the battery, tell an adult.



Never join + straight to – with nothing in between (a short circuit) – it gets hot fast.

machine 1

parking barrier

open & closed circuits



the idea – understand this first

A parking barrier is a car-park gate. It works as a **circuit** – a loop of battery, wires and a motor – and a **switch** opens and closes the loop. Break the loop anywhere and it stops. A torch is a simple circuit too.



the words to teach

circuit – the complete loop the electricity travels round

switch – opens and closes the loop, on purpose

current – the flow of electricity going round the loop

did you know (say it once): the flow of electricity going round and round the loop is called **current**.



the experiment + the answers

do: Build a loop: a power block, a switch and an LED. Turn the switch on so it lights. Now take one wire off the loop.

5-8: "With one wire off, the loop is broken. The light will:" → **stay off**

8-12: How many blocks does the current flow through on its way round the loop? → **all of them, in one path** (+ draw the schematic)



the circuit exercise

5-8 (complete it): "which block goes in the empty space, to turn the light on and off?" → **a switch block**

8-12 (design it): Design a circuit for a car-park gate: the guard can switch it ON and OFF, and a light shows when it is on. – *the light must only be on when the switch is on*



real life + the problem to solve

A torch, a doorbell and a table lamp are all simple circuits – a loop the electricity flows around.

problem: The parking barrier's light won't come on – one wire has come loose. Find the break and make the loop work again.

machine 2

wind turbine

polarity

**the idea – understand this first**

A wind turbine spins a **motor** from a battery to turn its blades and light an **LED**. Electricity has a **direction** (polarity): swap the wires and the motor reverses, and the LED only lights one way round.

**the words to teach**

motor – turns electricity into movement

led – a small light that only lights one way round

polarity – electricity's direction – the + side and the – side

did you know (say it once): a battery stores **charge** – like a store of energy, waiting to flow.

**the experiment + the answers**

do: Wire the motor to the power block so it spins. Now swap the two wires over.

5-8: "When you swap the two wires, the motor spins:" → **the other way**

8-12: Swapping the wires makes the motor spin: → **the opposite way** (+ draw the schematic)

**the circuit exercise**

5-8 (complete it): draw task – these three blocks are not joined up. draw the wires to make ONE loop, so power flows all the way round.

8-12 (design it): Design a circuit where a switch lets you choose WHICH WAY the motor spins – forward or backward. – *swapping the direction must not need any rewiring*

**real life + the problem to solve**

Batteries in a remote must go the right way round, or nothing works – that's polarity. problem: The wind turbine's motor is spinning the wrong way. Change one thing so it spins the way you want.

machine 3

soccer bot

polarity reversal

**the idea – understand this first**

A soccer bot drives and pushes a ball. A **direction switch** reverses a motor with no rewiring; two motors let it steer.

**the words to teach**

direction switch – flips which way current goes through the motor, so it reverses

reverse – to make the motor spin the opposite way

steer – to turn by driving the two motors differently

did you know (say it once): electricity flows from the + side all the way round the loop to the – side.

**the experiment + the answers**

do: Wire two motors through a direction switch. Drive forward, then flip the switch.

5-8: “Flipping the direction switch makes the bot:” → reverse

8-12: To make the bot TURN (not just go straight), the two motors must spin: → **opposite ways** (+ draw the schematic)

**the circuit exercise**

5-8 (complete it): “which block lets you flip the motors forwards and backwards?” → **a direction switch**

8-12 (design it): Design a circuit that drives TWO motors and can reverse both together, so the bot can go forward, back and turn. – *one control should flip both motors*

**real life + the problem to solve**

A remote-control car changes direction the same way – a switch flips the motor.

problem: The soccer bot only drives straight – it won't turn. Change one thing so it can steer.

machine 4

cleaning bot

one battery, many jobs

**the idea – understand this first**

A cleaning bot runs **three motors** from **one battery**. Sharing one path is a **series** circuit; each with its own path is **parallel**. A speed dial changes how fast they spin.

**the words to teach**

series – parts joined in one shared path

parallel – each part given its own path from the battery

speed – how fast a motor spins – changed with the speed dial

did you know (say it once): **voltage** is the push behind the flow – more push makes the motors spin faster.

**the experiment + the answers**

do: Run two motors in one path (series), then give each its own path (parallel). Watch their speed.

5-8: "Two motors sharing ONE path (series) each run:" → a bit slower

8-12: The motors run FASTER when they are wired in: → **parallel (each its own path)** (+ draw the schematic)

**the circuit exercise**

5-8 (complete it): draw task – draw the wires so BOTH motors get their OWN path from the battery – that makes a parallel circuit.

8-12 (design it): Design a circuit that runs THREE motors from ONE battery, each getting full power, with a single switch to turn it all off. – *full power to each motor means a parallel circuit*

**real life + the problem to solve**

A robot vacuum runs its wheels and brush from one battery, just like your cleaning bot. problem: One of the cleaning bot's motors won't run; the others are fine. Find out why and fix it.

machine 5

sensor-controlled crane

input & output



the idea – understand this first

A sensor crane stops on its own. A **sensor** is the **input**, the **motor** is the **output**, and a **motor driver** decides between them.



the words to teach

sensor – the robot's eyes – it senses something and sends a signal

input – what the machine senses (the sensor)

output – what the machine does (the motor spins, the led lights)

did you know (say it once): a **sensor** turns what it sees into a tiny electric signal the driver can read.



the experiment + the answers

do: Wire an IR sensor to the motor driver to the motor. Wave your hand in front of the sensor.

5-8: "The sensor sees your hand and tells the driver to:" → **change the motor**

8-12: The sensor makes the motor react when the object is: → **close enough (the trigger point)** (+ draw the schematic)



the circuit exercise

5-8 (complete it): "which block is the INPUT – the part that senses and starts it off?" → **a sensor block**

8-12 (design it): Design a circuit where a SENSOR stops the motor by itself when something comes near. – *show the path: input → driver → output*



real life + the problem to solve

Automatic taps, doors and night-lights all sense and respond – input to output.

problem: The crane doesn't stop when your hand is near. Change one thing so the sensor makes it stop.